Arab J Geosci
DOI 10.1007/s12517-013-0855-z

ORIGINAL PAPER

Characteristics, modification and environmental application of Yemen's natural bentonite

Aref Alshameri · Alkhafaji R.Abood · Chunjie Yan · Akhtar Malik Muhammad

Received: 17 September 2012 / Accepted: 21 January 2013

Abstract The bentonite deposit of Lahij Province, Yemen, has very promising commercial applications due to its mineralogy and physical and chemical properties. It was examined to determine its mineralogical composition, chemical and physical properties of the bentonite deposit, purity and sodium-exchanged bentonite. Modified bentonite was synthesized by exchanging cetyltrimethylammonium cations for inorganic ions on the bentonite and its adsorption properties for ammonium were characterized in batch experiments. Analytical methods were carried out to study the bentonite comprising X-ray diffraction (XRD), scanning electron microscopy (SEM), infrared spectroscopy, chemical analysis and kinetic and isotherm models were also tested. The results have shown that the purification of bentonite resulted in a bentonite fractions of the total sample composed of montmorillonite and <5 % quartz. The XRD data showed that the interlayer spacing (d_{001}) of bentonite decreased from 15.3 to 12.5 Å and then increased to 19.7 Å. Moreover, high cation exchange capacity, good water absorption and high swelling capacity were also obtained. The results have shown that the modified bentonite was more effective than the natural bentonite for ammonium removal. In addition to that, pseudo-second-order kinetic model, Freundlich and the Langmuir models described the adsorption kinetics and isotherm well. It was concluded that Yemen (Alaslef) bentonite can be potential adsorbents for ammonium removal.

Keywords Yemen's bentonite · Cetyltrimethylammonium · Bentonite characteristic · Adsorption kinetics · Isotherm

A. Alshameri · C. Yan (✉)
Engineering Research Center of Nano-Geomaterials of Education Ministry, China University of Geosciences, Lu Mo Road 388, Wuhan 430074, People's Republic of China
e-mail: chjyan2005@126.com

A. R.Abood
State Key Laboratory Of Biogeology and Environmental Geology, China university of Geosciences, Wuhan 430074, People's Republic of China

A. M. Muhammad
School of Environmental Studies, China University of Geosciences, Wuhan 430074, People's Republic of China

Introduction

The discovery of natural clay deposits has led to an increasing use of these minerals for the purpose of industrial and environmental applications (Kohno et al. 2009; Laredj et al. 2012). At present, bentonite is one of the most important clay-based minerals used in polymer processing. It can be used as a filler of nanocomposites showing unique mechanical properties (Zilg et al. 2001; Alexandre and Dubois 2000). Modified bentonite also exhibits high ability to remove hydrophobic contaminants or at least reducing, many long-standing pollution problems (Ghafari and Khezri 2012; Mohamed 2012) from aqueous solutions and therefore it is a very promising agent in environmental control (Anirudhan and Ramachandran 2007; Su et al. 2011; Kohno et al. 2009; Bhattacharyya and Gupta 2008; Crini 2006; Maicaneanu et al. 2009).

The use of clay minerals for environmental applications is gaining new research interests mainly due to their properties, significant worldwide occurrence and the simplicity of its method modification. The isomorphous substitutions of Al^{3+} for Si^{4+} in the tetrahedral sheet and Mg^{2+} or Fe^{2+} for Al^{3+} in the octahedral sheets result in a negative surface charge on the clay. This unbalanced charge is compensated by exchangeable cations (most often, Na^{+} or Ca^{2+}) which are relatively loosely held, although stoichiometrically, and give rise to major cation exchange properties. A second negative charge occurs as a result of dissociation of the hydrogen in various hydroxyl groups which include SiOH and AlOH groups. The layered structure of the clay allows

Published online: 07 February 2013

expansion (swelling) when contacted with water, which exposes an additional mineral surface capable of cation adsorption. Ammonium is one waste water contaminant that needs to be removed due to health concerns especially for wastewater (Liao et al. 2013; Farhadinejad et al. 2012).

Various methods including air stripping, biological methods and activated carbon have been used for ammonium removal. However, since biological methods do not respond well to shock loads of ammonium, unacceptable peaks over the discharging levels may frequently appear in the effluent $NH^{+}{}_{4}$ concentrations. However, high costs and poor regeneration are incurred in construction of the above methods and there is a higher risk to safety during the subsequent processing (Liao et al. 2012a). Compared with the above-mentioned methods, high safety, low cost and relative simplicity of application and operation are the attributes that are attracting resulted in increasing the focus on the use of bentonite in water-purification processes (Karadag et al. 2006; Saltali et al. 2007; Hedstrom and Amofah 2008). In the first step, natural bentonite needs to be modified to improve its sorption properties as well as purity before it can be used to remove ammonium effectively. Traditionally, some applications are based on the surface modified by the exchange of organic cations for inorganic ions on the layer surfaces of bentonite (Zampori et al. 2008; Majdan et al. 2008). As a result, organobentonites are powerful sorbents for nonionic organic pollutants relative to natural bentonite and other clays (Seredych et al. 2008; Abollino et al. 2003; Yan et al. 2008). In recent years, natural bentonite and its modified forms have also been reported for removal of ammonium from water systems but it seems that natural bentonite from various places have different characters. However, this research is unique because it is the first time to conduct detailed investigation and analytical research about bentonite of Yemen. As far as the removal of ammonium is concerned, there is no research that has been conducted focusing on adsorption isotherm and kinetics using Yemen bentonite.

The aim of this research lies in the fact that a special attention was paid to examine the characterization of Yemen bentonite and to evaluate its use for the removal of ammonium from aqueous solutions. The specific objective of this paper was to study the modification of bentonite samples (CATB-bentonite) and its use as an adsorbent to remove ammonium from aqueous solution. The effects of pH, stirring times and adsorbent dosages on NH_4^{+}removal of both the natural and modified samples was investigated and compared. Kinetics and isotherms of adsorption of ammonium were also investigated. Furthermore, for the purpose of characterizing the modified and natural bentonite, SEM, X-ray diffraction (XRD) and FTIR spectroscopic techniques were conducted.

Geological setting

Description of the study area

Alaslaf area is located about 7 km northeastern of Alrahidah (Alqbitah) in the Lahij province south Yemen, which extends to 0429120 E and 1479830 N (Fig. 1a).The region is covered by basic rocks (Migmatite–Gneiss) overlained by Triassic volcanic (Khaled and Philippe 2010), represented by acidic rocks (Rhyolite–Tuff– Ignimbrite), basaltic rocks and quaternary deposits as shown in Fig. 1b.

Lithological characteristics

The clay minerals in this region is greenish gray Rhyolitic tuff of Triassic age, which seems to be decomposed to fine-grained friable soft mud in its outer rims with soapy texture (Bentonite), some of them contains pieces of volcanic glass. The outcrops of clay minerals are represented by greenish gray Rhyolite tuff of tertiary. The external parts of these rocks are decomposed to fine-grained friable soft-clay materials with soapy texture (bentonite); some of them contain pieces of black volcanic glasses (pitchstone), zeolite and a small amount of iron oxides. While their internal parts appear as rock fragments (fragile to semi-coherent) as contained in some outcrops streaks 1–2 cm of decomposed volcanic glass.

In general, the apparent thickness of these outcrops ranged between 2 m in Alaslaf outcrop and 10 m in Warzan-Simda valley. The extensions of these outcrops are also different from one place to another, 5–15 m in Alaslaf and reaches 70 m in Warzan-Simda valley. It has not been possible to follow up some outcrops because they were covered by rocks debris which resulted from weathering processes. The clay outcrops are overlaid by tuffaceous ignimbrites with a thickness ranging between 1.5 m in Alaslaf and 15 m in Warzan–Simda valley. These clay layers are laid over reddish brown partially decomposed tuff into sticky clay in some outcrops.

Experimental and methods

Materials and analytic methods

A natural bentonite (NB) from Alaslef-AlQbitah in Lahij province, Yemen, was used for this research. Phenolphthalein, sodium chlorite, magnesiumoxid, ethanolabsolute, sodiumhydroxide, cetyltrimethyl-ammonium

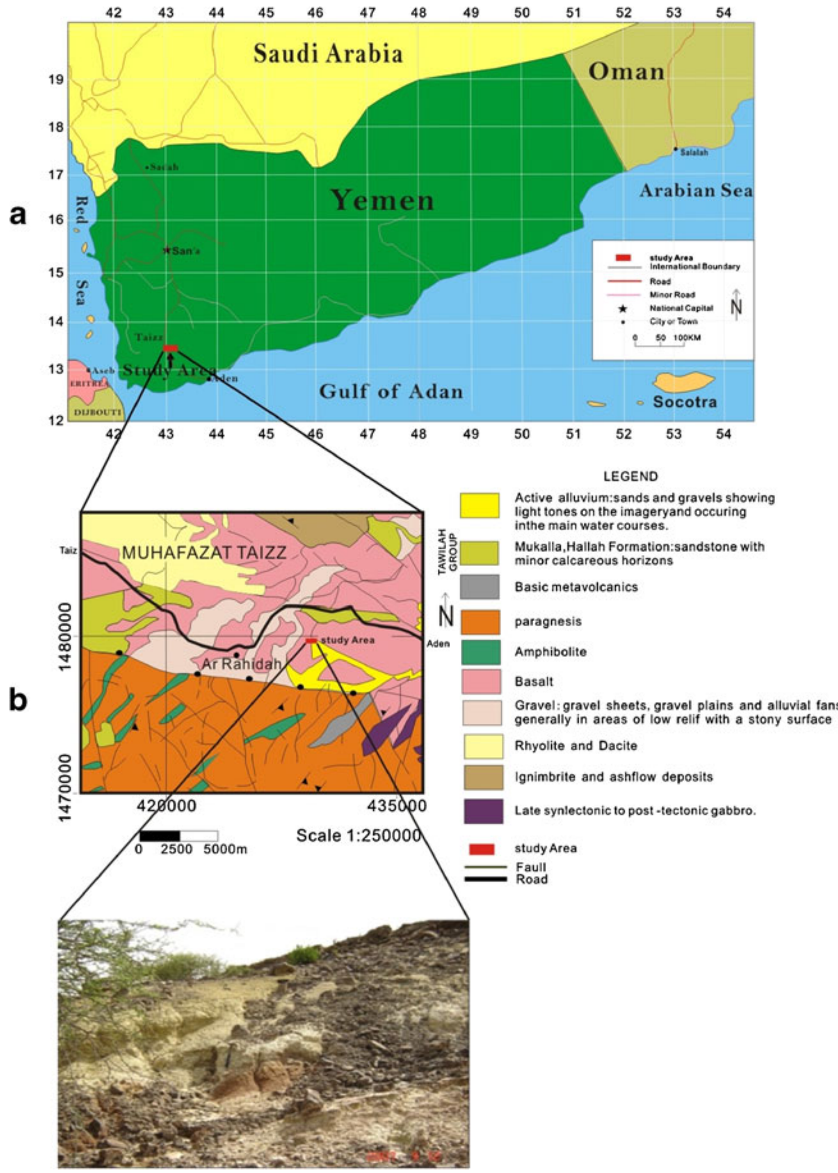

Fig. 1 **a** Location map and **b** geological map of study the area

bromide (CTAB), silver nitrate, hydrochloric acid, ammonia chloride, ammonia hydroxide, formaldehyde, potassium biphthalate, calcium chloride and other reagents were used for analytical grades

Preparation of samples

Sample purification The natural bentonite samples were crushed in a mortar and were placed in several 1-L graduated cylinders. The crushed samples were mixed with distilled water and left for 12 h. The water above was separated and transferred to other cylinders, a few drops of 0.1 M HCl were added and the samples were left for 8 h or more as deposits of bentonite settle on the bottom of the cylinder; the water was thrown then the solid precipitates were collected and dried at 60 °C for 24 h. This process was repeated several times.

Preparation of sodium-exchanged bentonite or Na-bentonite The solid phase was saturated with sodium ions by stirring in a 1 M sodium chloride solution for 12 h, the bentonite was separated from the suspension by centrifugation at 1,000 rpm/10 min. This process was repeated three times and the saturation was achieved. In order to eliminate excess chloride ions, the solid was washed with distilled water several times and a silver nitrate test was conducted to confirm the absence of Cl ions. This was dried at 50 °C and the material obtained is called sodium-exchanged bentonite or Na-bentonite.

Preparation of organobentonite Modified bentonite (MB) was synthesized by exchanging cetyltrimethyl ammonium cations for inorganic ions on the bentonite: 10 g of bentonite was mixed with 100 ml of 2 % CTAB solutions. The mixtures were subjected to mechanical stirring for 3 h in water bath at 80 °C. In this condition, the MB was obtained completely. The precipitate was separated by filter paper, washed with hot distilled water for several times until the removal of bromide ions from the sample as 0.1 mol/L $AgNO_3$ was used indicators. The sample was dried at 60 °C for 24 h and finely crushed.

NH_4^+ adsorption experiments

All adsorption in batch experiments were carried out of stopper conical flasks (500 ml) in magnetic stirring water bath and 120 rpm rate of bentonite/liquid ratio of 1 g/100 ml at 28 °C. A stock solution (1,000 mg/L) was prepared by dissolving 3.819 g NH_4Cl in 1.0 L distilled water; desired concentrations were obtained when needed by diluting the stock solution with distilled water. For determining the optimum time required of ammonium removal by NB and MB from aqueous solution to reach equilibrium, a weighed quantity of adsorbents (1 g) was added into solution of 20 mg/L ammonium concentration and was stirred for 10–80 min of both bentonites at a fixed pH7 and temperature 28 °C. In the adsorption, experiment was conducted using various ammonium concentrations from 5 to 50 mg/L. For determining bentonite loading effect, bentonite loading was varied from 0.5 to 2.5 g/100 ml at initial ammonium concentration of 20 mg/L. The bentonite liquid was then stirring for 40 min of both bentonites at temperature 28 °C. To investigate the effect of pH, the adsorption at initial ammonium concentration of 20 mg/L was performed at different pH values (2–10) at 28 °C; batch adsorption was conducted for 40 min of both bentonites and the pH was adjusted by adding few drops of diluted HCl and/or NaOH solution. The suspension was filtered through a 0.45-μm membrane filter of. The residual concentration of ammonium was determined by Nessler's reagent spectrophotometry method at 420 nm. The removal efficiency (in percent) of bentonite and the amount of adsorbed ammonium ions (q_e) by it were calculated, respectively, using the following equations:

$$\text{Removal efficiency}(\%) = \frac{(C_0 - C_e)}{C_0} \times 100 \tag{1}$$

$$q_e = \frac{(C_0 - C_e)V}{m} \tag{2}$$

Where:C_0 (in milligramme per litre) represents the initial concentration of ammonium, C_e (in milligramme per litre) represents the equilibrium of ammonium concentration, V is the volume of solution (litre), q_e (in milligramme per gramme) is the amount of ammonium adsorbed and m (in gramme) is the mass of bentonite.

Characterization methods

Identification of mineral species in the bentonite samples was carried out by XRD analysis of the random-oriented powder samples using Philips χ'Pert PRO XRD. The X-ray tube was operated at 40 kv and 40 mA with Cu Kα radiation at a scanning rate of 5°/min for 2θ ranging from 0 to 65 of X-ray diffractometer. Morphological and chemical compositions and BET surface area of bentonite were carried out on Quanta 200 of SEM, atomic absorption spectroscopy and nitrogen adsorption at 77 K with an Automatic Volumetric Sorption Analyzer (ASAP2020, TSI, America), respectively.

The infrared absorption spectroscopy (IR) of the product was obtained with NICOLET AVATAR 370 DTGS FT spectrometer in the range of 400–4,000 cm^{-1} at room temperature and the samples were mixed with KBr salt. The cation exchange capacity (CEC), gelling value and swelling capacity were measured with a stander method GB/T 20793-2007 (200) and water absorption was measured with physical and chemical properties testing of non-metallic methods was evaluated by stander method (GB/T 20793–2007). The residual concentration of ammonium was determined by a standard method (water quality determination of ammonia nitrogen—Nessler's reagent spectrophotometry method, GB 7479–87), the absorbance at a wavelength of 420 nm with UV–visible recording spectrophotometer (U3210,Hitachi, Tokyo, Japan) using 10-mm matched quartz cells. Quality control testing includes experiments with blanks and duplicates. The average data are presented.

Result and discussion

Characterization of natural and modified bentonite

XRD (Fig. 2a) shows that the natural bentonite of Alaslef-Alqabbaitah area is composed of montmorillonite and the major impurities are quartz, gypsum, pyrophyllite, orthoclase, calcite and albite. Conversely, it shows that the total purified bentonite is composed of montmorillonite with a small amount of Quartz (<5 %) as shown in Fig. 2a.

Table 1 depicts a semiquantitative mineralogical composition. It can be seen that the bentonite sample was composed of 95 % montmorillonite with less quartz (<5 %) as an impurity of purified bentonite. While the natural bentonite composed of 50–60 % montmorillonite, 10–15 quartz, 5± gypsum, 15± albite, 5± calcite and 5± pyrophyllite. In comparison with the mineralogical composition of purified Yemen bentonite with purified Wyoming and Tunisian

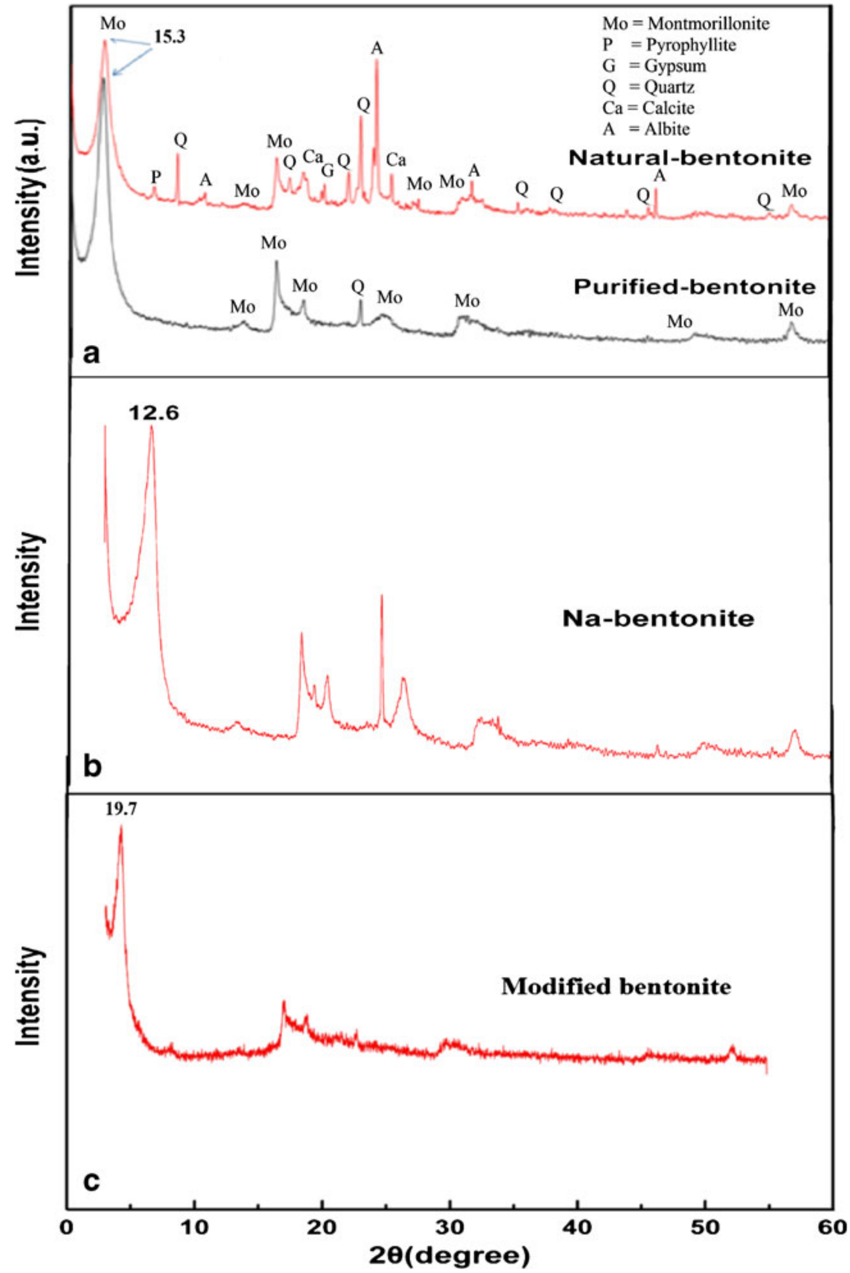

Fig. 2 XRD of the **a** natural and purified bentonite, **b** Na-bentonite and **c** modified bentonite by CTAB

Table 1 Semi-quantitative mineralogical composition of studied bentonite, Wyoming and Tunisian bentonites (in purified and natural forms) for comparison [wt%]

Studied bentonite (Yemen bentonite)			Wyoming bentonite [a]		Tunisian bentonite [b]	
Mineral/sample	Natural	Purified	Natural	Purified	Natural	Purified
Montmorillonite	50–60 %	>95 %	85 %	100 %	60	88
Quartz	10–15	<5	10	–	5	–
Gypsum	5±	–	–	–	–	–
Albite	15±	–	–	–	–	–
Calcite	5±	–	5	–	15	–
Pyrophyllite	5±	–	–	–	–	–
Illite	–	–	–	–	10	12
zeolite	–	–	–	–	10	–

[a]Laribi et al. (2006)

[b]Laribi et al. (2007)

bentonites (Laribi et al. 2006, 2007), they are comparable as given in Table 1.

Figure 2b shows that the decrease of the basal space from 15.3 to 12.6 Å indicated the presence of exchangeable Na^+. It thus represents Na-exchanged bentonite. In the case of the modified bentonite, d_{001} value was determined approximately to be 19.7 Å, which identifies the structural configuration of alkyl chains in the interlamellar space which is pseudotrimolecular layer (19.7 Å) as shown in Fig. 2c. Then, the modified bentonite was confirmed by IR analysis and SEM image as given in Figs. 3 and 4.

Figure 3 illustrates that the strong absorption peak occurs at 1,484 cm^{-1} which can be related to bending vibrations of NH_4^+ and their bonding supporting the modification of surfactant molecules between the silica layers. The attributions of absorption at 988 cm^{-1} can be due to Si–O–Si and Si–O–Al stretching 885 cm^{-1} which can be due to the deformation of Si–OH. Normally, the absorption-related stretching vibrations of Si–O–Si and Si-O-Al occur at 1,100–1,000 cm^{-1}, but this absorption is changed for lower wave number after modification of bentonite.

Atypical "corn flakes" texture is manifested by using SEM, which identified the characteristics of natural bentonite as can be seen in Fig. 4a, b, which is characterized by good equalisation, the particles are formed of irregular shape. In contrast, modified bentonite is considerably different from the parent bentonite as shown in Fig. 4c, d.

Chemical data of the bentonite in both natural and modified bentonite have shown that the bentonite from Yemen (Alaslef-Alqabitah) is composed mainly of SiO_2 and Al_2O_3 with the other oxides being present in trace amounts as given in Table 2, which reflect the mineralogy of bentonite sample (Table 1). The chemical composition of Alaslef-Alqabitah's bentonite samples in purified bentonite is comparable with bentonite deposits of high economic value (Laribi et al. 2006, 2007) as presented in Table 2.

The CEC, water absorption, gelling value and swelling capacity were measured and they are 129.94 mmol/100 g, 202 %, 98 ml/15 g and 32.5 ml/g, respectively. The results have indicated that the bentonite of Alaslef-Alqabitah area is characterized by high CEC, good water absorption and high swelling capacity.

BET-specific surface area values of the samples are given in Table 3. It can be clearly seen that there was a decrease in the order of modified sample. By contrast, the average pore diameter of MB is higher than that of the NB. This may be due to the blocking of the micro pore openings on the surface by large organic criterions or formation of macroporous structure.

Adsorption kinetics

A weighed quantity of NB and MB, respectively, were added into 200 ml of ammonium solution (the concentration was 20 mg/L), with solid/liquid ratio of 1 g/100 ml, at 28°C, adjust adsorption time and the results were shown in Fig. 5. As shown in Fig. 5, the removal ratio of ammonium ion increased with increasing stirring time and 30 and 55 % of ammonium ion removal was completed within 10 min of NB and MB, respectively. This also confirmed larger adsorption capacity of MB to NB, it might be because of better structure and properties of MB to NB. The experiment data showed that MB could reach high removal (>78 %) in 40 min and after that, the removal efficiency did not change considerably; as to NB, the ammonium removal rate reaches high (35 %) in 40 min. Therefore, the optimum value of stirring time was found to be 40 min for both bentonites. It reached up 35 and 80 % within 40 min of NB and MB, respectively, and became very slow with increasing of stirring time. This may be ascribed to the utilization of the most readily available adsorbing sites of the bentonite that leads fast diffusion and rapid equilibrium attain (Karadag et al. 2006).

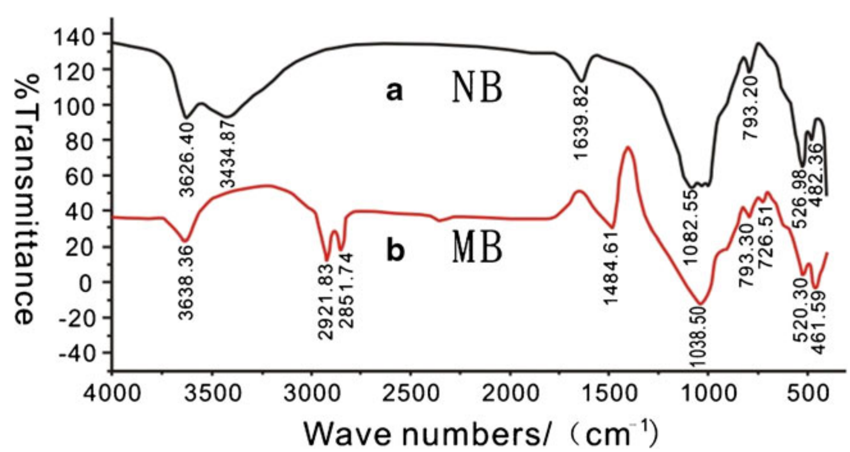

Fig. 3 IR spectra of (*a*) the natural bentonite (*NB*) and (*b*) modified bentonite (*MB*)

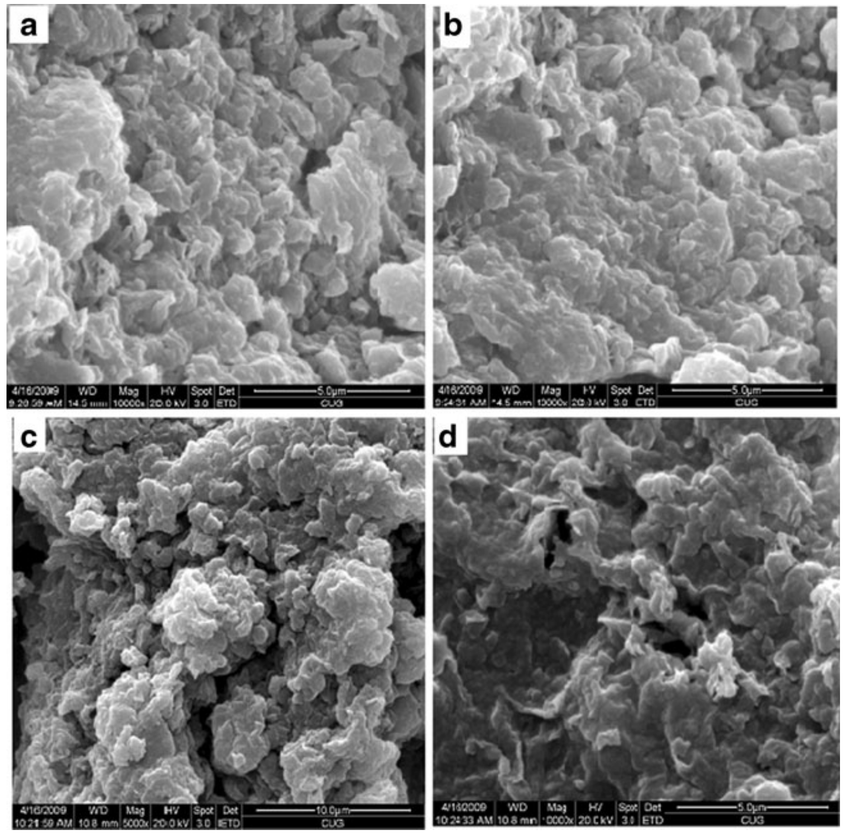

Fig. 4 SEM of the natural bentonite (NB) (**a**, **b**) and modified bentonite (MB) (**c**, **d**)

To evaluate the differences in adsorption kinetics between adsorbents, ammonium adsorption was fitted using a pseudo-first-order, pseud-second-order kinetic and intraparticle diffusion models to describe the adsorption process as follows.

Pseudo-first-order kinetic model It is expressed using the following equation:

$$\ln(q_e - q_t) = \ln q_e - k_1 t \tag{3}$$

Pseudo-second-order kinetic model It is expressed using the following equation:

$$\frac{dq_t}{dt} = k(q_e - q_t)^2 \tag{4}$$

Where q_t (in milligramme per gramme) is the adsorbed amount at time t, k_1(in per minute) and k (in milligramme per gramme minute) are the pseudo-first-order and pseudp-second-order rate constant, respectively, and q_e (milligramme per gramme) is the adsorbed amount at equilibrium.

Table 2 Chemical composition of the analyzed samples (wt%) of studied bentonite, Wyoming and Tunisian bentonite (in purified and natural forms) for comparison

	Studied bentonite (Yemen bentonite)		Wyoming bentonite[a]		Tunisian bentonite[b]	
Oxide/sample	Natural	Purified	Purified	Natural	Purified	Natural
SiO_2	62.12	54.18	56	49	57	56
Al_2O_3	14.53	15.55	18	17	16	11
Fe_2O_3	4.25	5.53	4.1	4.5	5.0	4.4
TiO_2	0.45	0.59	0.58	0.03	2.6	3.3
MgO	1.81	1.48	1.5	3.7	0.51	0.59
Na_2O	1.88	2.80	3.2	2.7	3.2	2.7
CaO	1.59	0.74	1.8	3.1	3.6	5.7
K_2O	0.74	0.30	0.24	0.84	1.1	1.3
Loss on ignition	13.03	19.06	11	19	11	15

[a] Laribi et al. (2006)

[b] Laribi et al. (2007)

Table 3 BET surface area results

Adsorbents	BET surface area m^2/g	Total Volume (mL/g)	Average pore diameter (Å)
Natural bentonite	32.5	0.0033	72.55
Modified bentonite	8.5	0.0004	170.9

Integrating (4) for the boundary condition t=0 to t=t and q_t=0 to q_t=q_t, it was rearranged to obtain a linear form:

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{1}{q_e} t \tag{5}$$

Intra-particle diffusion It is expressed by using the following relation:

$$\text{Weber} - \text{Morriss Eq.}: \quad q_t = k_{id} t^{1/2} + C \tag{6}$$

Equation 6 is an empirically found functional relationship, where uptake varies almost proportionally with $t^{1/2}$ rather than with the contact time t, k_{id} is the intraparticle diffusion rate (in milligramme per gramme root square minute) obtained from the slope of the straight line of q_t versus $t^{1/2}$.

The adsorption kinetics of ammonium by MB and NB are presented in Fig. 6. The kinetic data were better fitted by the pseudo-second-order model than by the pseudo-first-order model (Table 4). The pseudo-second-order model indicates that chemisorption dominated in the adsorption process (Vimonses et al. 2009). The difference in the adsorbed concentration of adsorbate at equilibrium (q_e) and at time t (q_t) is the key driving force for the adsorption, and the adsorption capacity is proportional to the number of active adsorption sites occupied on the adsorbent (Shen et al. 2009; Wang et al. 2011). There are three steps involved in pseudo-second-order kinetic model: (1) the ammonium molecules diffuse from liquid phase to liquid–solid interface; (2) the ammonium molecules move from liquid–solid interface to solid surfaces; and (3) the ammonium molecules diffuse into

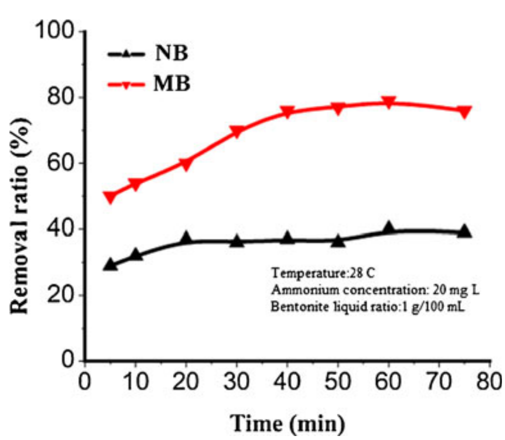

Fig. 5 Adsorption kinetics of ammonium by *NB* and *MB*

the particle pores (Wang et al. 2011; Liao et al. 2012b). Herein, the diffusion of ammonium molecules from aqueous phase was much faster than the surface and intraparticle diffusion processes (Liao et al. 2013). To reveal the relative contribution of surface and intraparticle diffusion to the kinetic process, the kinetic adsorption data were further fitted with the Weber-Morris model.

According to this model using Eq. (7), the plot of uptake, q_t, versus the square root of time (t1/2) should be linear if intraparticle diffusion is involved in the adsorption process. As can be seen from Fig. 6c, ammonium exchange by both bentonites involves two stages. These two stages suggest that the ammonium exchange process proceeds by surface sorption and intraparticle diffusion. The first one can be attributed to external surface adsorption or an instantaneous adsorption stage, where the ammonium molecules are transported to the external surface through film diffusion and its rate is very fast. The second region is due to a gradual adsorption stage, where the ammonium molecules are entered into modified bentonite particle by intraparticle diffusion (Kumar et al. 2005). The values of intercept C provide information about the thickness of the boundary layer, the resistance to the external mass transfer increases as the intercept increases. The constant C was found to increase from 0.0736 to 0.1376 mg/g of NB to MB as shown in Table. 4, which indicates the increase of the thickness of the boundary layer and decrease of the chance of the external mass transfer and hence increase of the chance of internal mass transfer (Nemr et al. 2009). Table 4 presents the results of fitting experimental data to the pseudo-first-order, pseudo-second-order and intraparticle models. It can be seen from Table 4 that the correlation coefficient R^2 varies in the order: pseudo-second-order > pseudo-first-order > intraparticle diffusion model under all experimental conditions, which indicates that the pseudo-second-order model is the most suitable in describing the adsorption kinetics of ammonium on bentonite.

Adsorption isotherms

Adsorption isotherm are essential for the description of how NH_4^+ concentration will interact with bentonite and are useful to optimize the use of bentonite as adsorbents. For adsorption isotherms, Langmuir model and Freundlich model were employed. Langmuir model is valid for monolayer adsorption onto a surface with finite number of identical sites. Langmuir parameters were determined by the following formula:

$$q = \frac{q_{max} K C_e}{1 + K C_e} \tag{7}$$

Where q_{max} (in milligramme per litre) and K (in litre per milligramme) are the monolayer capacity attained at high concentrations and the equilibrium constant, respectively. C_e

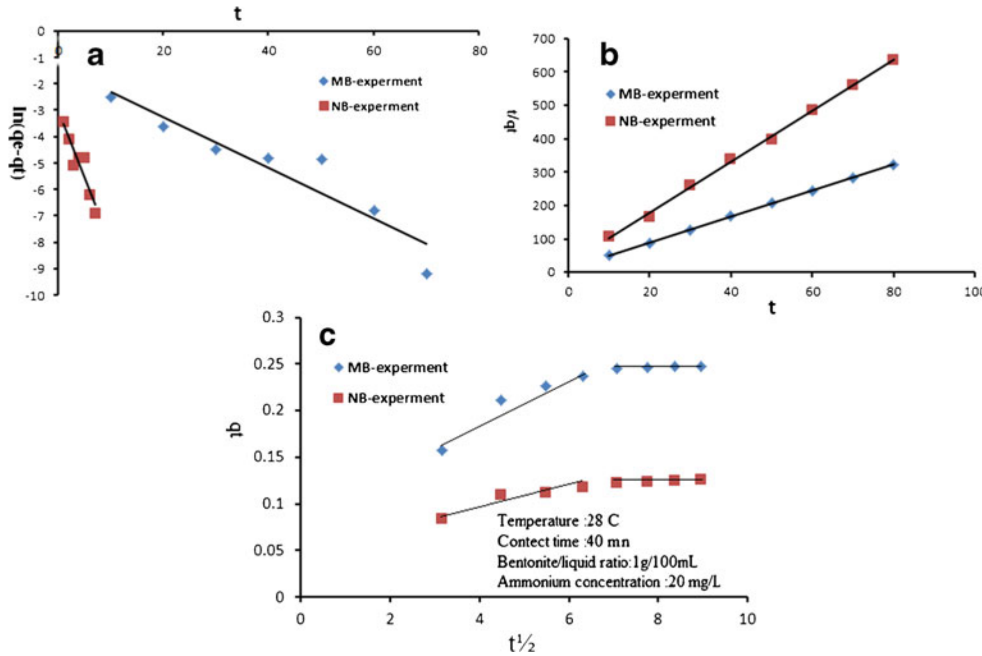

Fig. 6 **a** Pseudo-first-order, **b** pseudo-second-order, **c** intraparticle diffusion kinetic plots for adsorption of ammonium on NB and MB

is the equilibrium concentration in solution (in milligramme per litre) and q represents the amount adsorbed at equilibrium (in milligramme per gramme). The linearisation of Eq. (7) is given by Eq. (8) for Langmuir data fitting.

Table 4 Pseudo-first-order, pseudo-second -order, intraparticle diffusion model kinetic, Langmuir and Freundlich parameters and regression coefficients

	Natural bentonite (NB)	Modified bentonite (MB)
Pseudo-first-order kinetic parameters		
q_e	0.1258	0.2478
K_1	0.5087	0.0956
R^2	0.8685	0.8802
Pseudo-second-order kinetic parameters		
q_e	0.01238	0.02467
K_2	607.4	109.52
R^2	0.9924	0.9995
Intraparticle diffusion model		
K_{id}	0.0064	0.0139
C	0.0736	0.1376
R^2	0.8528	0.8338
Langmuir model		
q_{max}	0.0698	0.0967
K	0.5537	0.7129
R^2	0.9634	0.9971
Freundlich model		
k_F	0.1044	0.2152
n	1.408	0.8298
R^2	0.9097	0.9304

$$\frac{C_e}{q} = \frac{1}{kq_{max}} + \frac{1}{q_{max}} C_e \tag{8}$$

Furthermore, Freundlich model assumes a heterogeneous surface with a non-uniform distribution of heat of adsorption over the surface and binding sites are not equivalent and/or independent. Freundlich parameters were determined by the formula:

$$q = k_F C_e^{\frac{1}{n}} \tag{9}$$

Where k_F and $1/n$ represent the Freundlich capacity factor and the Freundlich intensity parameter, respectively. C_e is the equilibrium concentration in the solution (in milligramme per litre) and q represents the amount adsorbed at equilibrium (in milligramme per gramme). The linearisation of Eq. (9) is given by Eq. (10) for Freundlich data fitting:

$$\log q = \log k_F + \frac{1}{n} \log C_e \tag{10}$$

Langmuir model The linear plot of Langmuir isotherm of MB and NB is shown in Fig. 7a. It is noted that the values of q_{max} and k were calculated from the slope and the intercept of the plot using Eq. (8) and are given in Table 4. It will be seen that applicability of the simple Langmuir equation for the present isotherm data indicates that the Langmuir equation was able to describe properly the isotherm of ammonium on two bentonites (correlations coefficient >0.95). As shown in Table 4, the MB had much higher ion exchange capacity than NB.

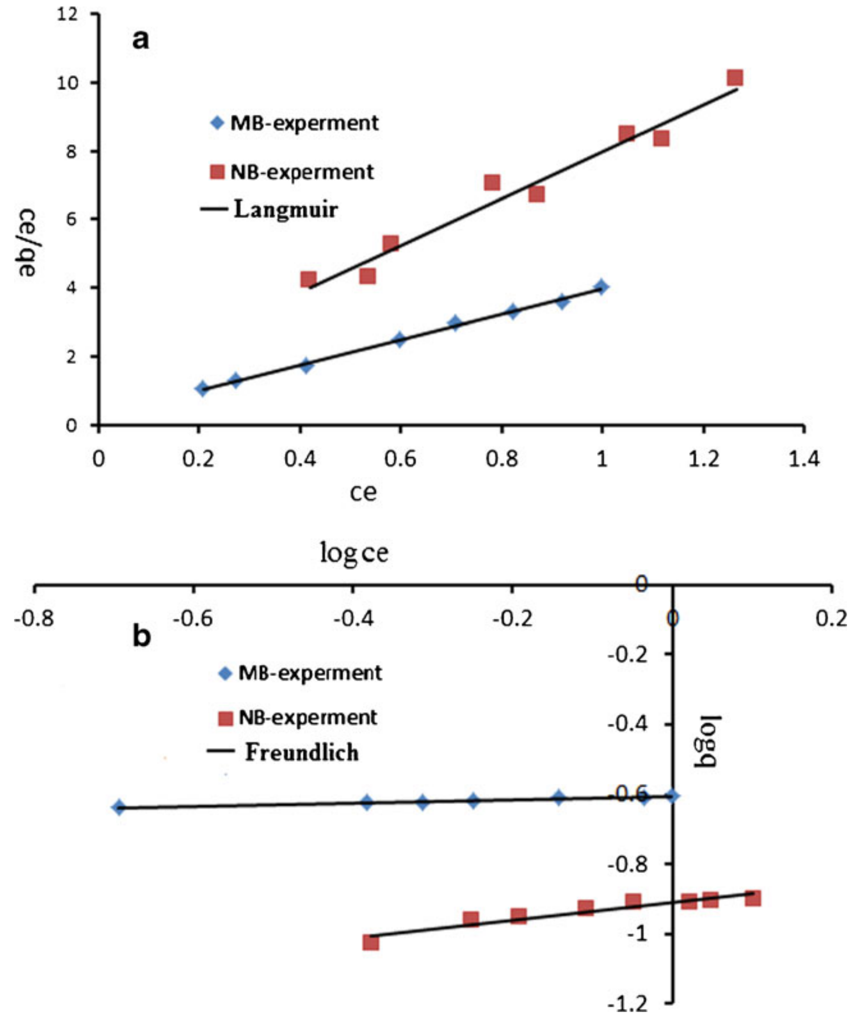

Fig. 7 The adsorption isotherms of ammonium on both bentonites **a** Langmuir and **b** Freundlich

Freundlich model Figure 7b shows the linearised Freundlich adsorption isotherm of ammonium curve and the Freundlich parameters is presented in Table 4. The fit of the data for ammonium adsorption onto MB and NB suggests that the MB gives better fittings than NB of Freundlich model, as is obvious from a comparison of R^2 in Table 4.

A summary of the models used and the coefficients which are calculated by experimental data and Eqs. (8) and (10) are given in Table 4. It is seen that applicability of the simple Langmuir and Freundlich equation for the present isotherm data indicates that the Freundlich and Langmuir equations was able to describe properly the isotherm of ammonium on two bentonite (correlations coefficient >0.90). In Langmuir equation, coefficient k represents the maximum amount that can be sorbed. Furthermore, k_F is Freundlich constant that is an index to the sorption capacity of sorbent. Both the values for k and k_F indicated that the modified bentonite has the higher sorption capacity than the natural bentonite.

The ammonium removal on the bentonite could be mainly attributed to the fact that the bentonite is composed of aluminosilicate tetrahedral, where substitution of each aluminum (Al^{3+}) for silicon (Si^{4+}) provides a negative charge on the bentonite framework (Englert and Rubio 2005). The negative charge is balanced by positively charged ions such as Na^+, K^+, Ca^{2+} and Mg^{2+} on the aluminosilicate structure with weaker electrostatic bonds (Rozic et al. 2000). As a result, those cations in the bentonite structure can be easily exchanged with other cations in the solution such as ammonium (Farkas et al. 2005).

Effect of adsorbent dosage

Figure 8 shows that the removal efficiency of ammonium ions by two bentonites increased with increasing the amount of both bentonites due to the increase of total available surface area and therefore the absorbent sites of the bentonite (Widiastuti et al. 2011). As shown in Fig. 8, the ammonium removal rate of the MB was higher than that of NB at each adsorbent dosage and the appropriate dosage of modified adsorbent was 1.5 g (Fig. 8), the ammonium removal rate was 80 % and it changed a little with an increase of dosage.

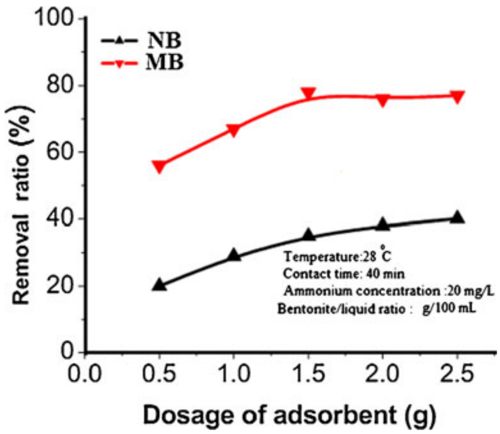

Fig. 8 Effect of bentonite dosage on ammonium adsorption

Hence, when the ammonium exchanged completely with cations on the bentonite surface at a certain amount of bentonite loading, the ammonium removal reached equilibrium.

Effect of the solution pH

Figure 9 shows that there were not some differences in suitable pH ranges of NB and MB. The removal efficiency of both bentonites increased by raising the pH from 1 to 6 and reached a maximum value at pH6, and then it decreased at pH values 8, 9 and 10. This result is consistent with those reported by Du et al. (2005) who found that the optimum ammonium removal efficiency was achieved at pH6. Conversely, Widiastuti et al. (2011) reported that zeolite was more selective at pH5. However, variation in bentonite and zeolite performance in the ammonium removal efficiency at various pH is due to the change in ammonium behaviour in aqueous solutions at various pH. The ammonium in aqueous solution can be found as ammonium, NH_4^+, a dissociated form or as ammonia, NH_3, a non-dissociated form and their formation depend upon pH solution. A diagram of ammonia behaviour in aqueous solution was reported by Leyva-Ramos et al. (2004) who found that at pH lower than 7, the ammonium ion, NH_4^+, was a predominant species, while at pH greater than 10, the predominant species was NH_3. Rozic et al. (2000) reported that the ammonium bond to the clays and zeolite samples, ion exchange is the main mechanism of ammonium removal by natural bentonite besides adsorption mechanism. Based on this and the diagram reported by Leyva-Ramos et al. (2004), it can be explained that at pH5 to 7 where the ammonium ion NH_4^+ was a predominant species can incur ion-exchange with the cations in the bentonite. Therefore, at pH greater than 6 where the NH_3 was a predominant species, the ammonium removal efficiency was low due to the low ion-exchange ability of NH_3.

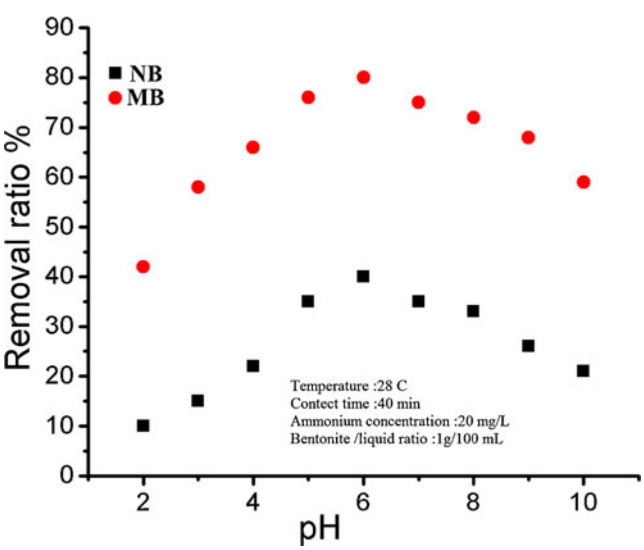

Fig. 9 Effect of pH on the removal of ammonium ion

Conclusions

This paper presents the characteristics of Yemen bentonite and its efficiency in removing ammonium. The characteristics and efficiency were based on the bentonite chemical composition and physical properties. Besides, sodium-exchanged, modified bentonite and adsorption behaviour of ammonium from aqueous solution onto bentonite were investigated. The following conclusions were reached by means of experimental results:

- The bentonite sample after the removal of impurities minerals resulted in a bentonite fractions of the total sample composed of montmorillonite and a small amount of quartz (<5 %).
- XRD shows that the interlayer spacing (d_{001}) of bentonite is decreased from 15.3 to 12.5 Å, indicating the presence of exchangeable Na^+ which is Na-bentonite and then modification of bentonite by CTAB surfactant expands the basal space to 19.7 Å.
- The mineralogical and chemical composition of Yemen's purified bentonite comparable with bentonite deposits of high economic value (Wyoming and Tunisian purified bentonite).
- The highest adsorption capacity was obtained at stirring time of 40 min for both bentonites and pH has a significant effect on the removal of ammonium by both bentonites. The highest adsorption capacity was obtained at pH6.
- The percent of ammonium removal increased with increase of amount of bentonite loading, and the appropriate dosage of modified adsorbent was 1.5 g, the ammonium removal rate was 80 %.
- Langmuir and Freundlich adsorption isotherm of both bentonites fit well with the equilibrium adsorption data; this adsorption process follows a pseudo-second-order kinetics rate model.

- The modification of bentonites by CTAB surfactant can remove ammonium efficiently; it had the higher removal efficiencies than that natural bentonite. The study shows that bentonite and modified bentonite can be used as cheap, efficient and eco-friendly adsorbent for removing ammonium from water and wastewaters. In addition, on the basis of bulk mineralogical composition, chemical and physical properties of Yemen bentonite indicate that purified bentonite could be potential raw materials for industrial applications. Further study is necessary to establish the process parameters for generation of better quality of products.

Acknowledgments A. Alshameri wishes to thank the Geological Survey and Mineral Resources Board, Sana'a, Yemen, and Engineering Research Center of Nano-Geomaterials of Education Ministry, China University of Geosciences, Wuhan for supporting this study. We are grateful to the anonymous reviewers for their valuable suggestions.

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